



TFG – Resolution Studies

Simulation and analysis tools

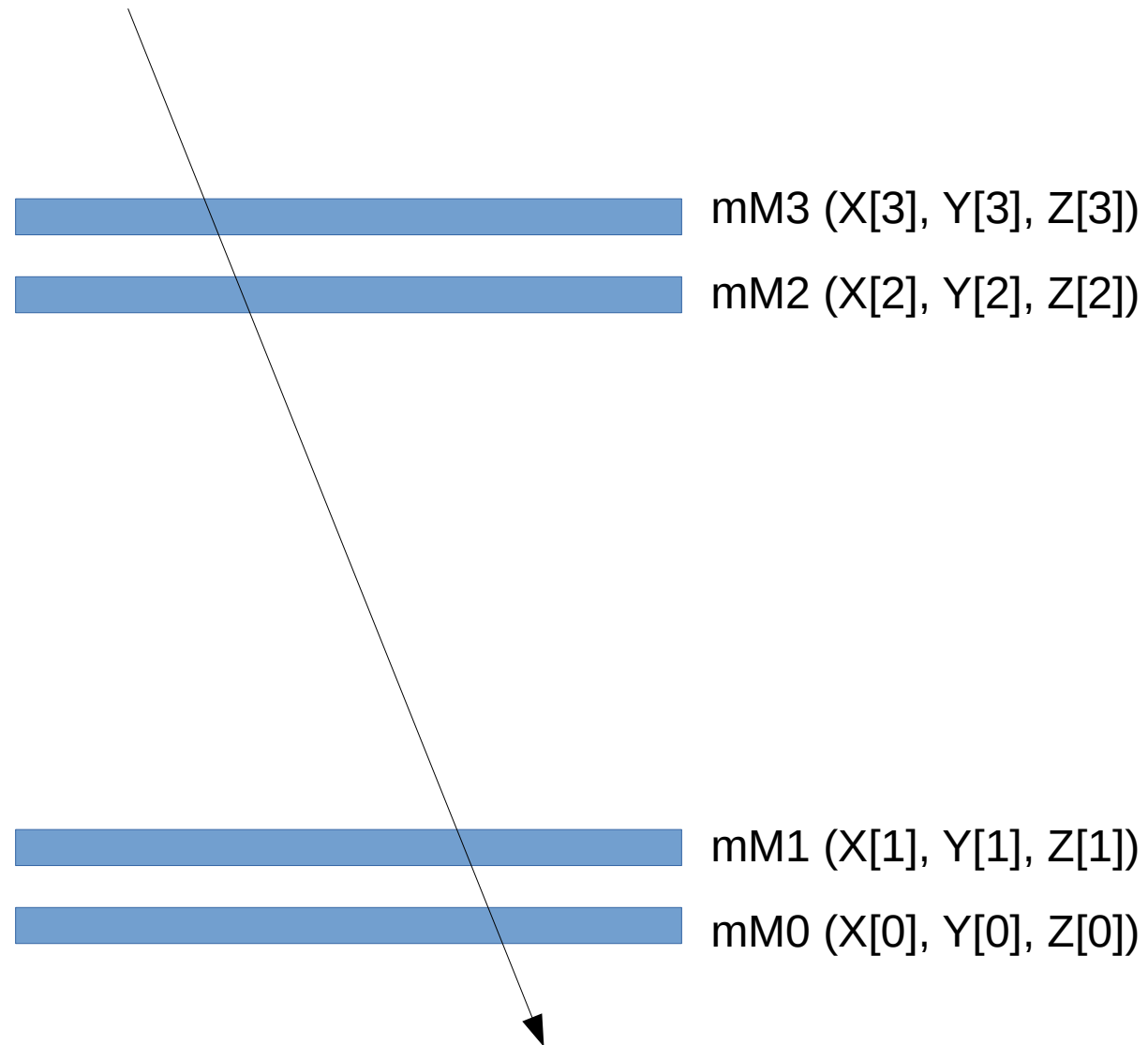
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Simulations – Geant4: G4TomoDet code
@ /home/hgomez/G4TomoDet_TFG2025

Output : G4TomoDet_*.root

en scripts: root

G4TomoDet_Data_Check_LightTree_Residual("../data/G4TomoDet_5E7_Guan_OpenSky_titan_slurm_111_160.root")

Esto lanza una simulacion creo, y me pone la posicion de los detectores

tfg_2025_pablo@sultan:~/G4TomoDet_TFG2025/src\$ cat DetectorConstruction.cc

// Chamber Detector

fChamberXSize = 50. *cm;

fChamberYSize = .8 *cm;

fChamberZSize = 50. *cm;

fGapBigChamber = 13. *cm;

fGapSmallChamber = 7.0 *cm;

// gap beetween chambers pairs

// gap beetween chambers of a pair

// Fermi P2

fchamberRotAngleX = 0.0*deg;

fchamberRotAngleY = 0.0*deg;

fchamberRotAngleZ = 0.0*deg;

Para nueva simulacion: cd ~/G4TomoDet_TFG2025

./build/G4TomoDet macros/muons.mac > nueva_simulacion.log

- All the simulations information:

en data: root -l --web=off -e 'new TBrowser();' NombreDelArchivo

- MetaData

- General aspects

- Detector geometry...

- Muons

me saca la geometría que defino:

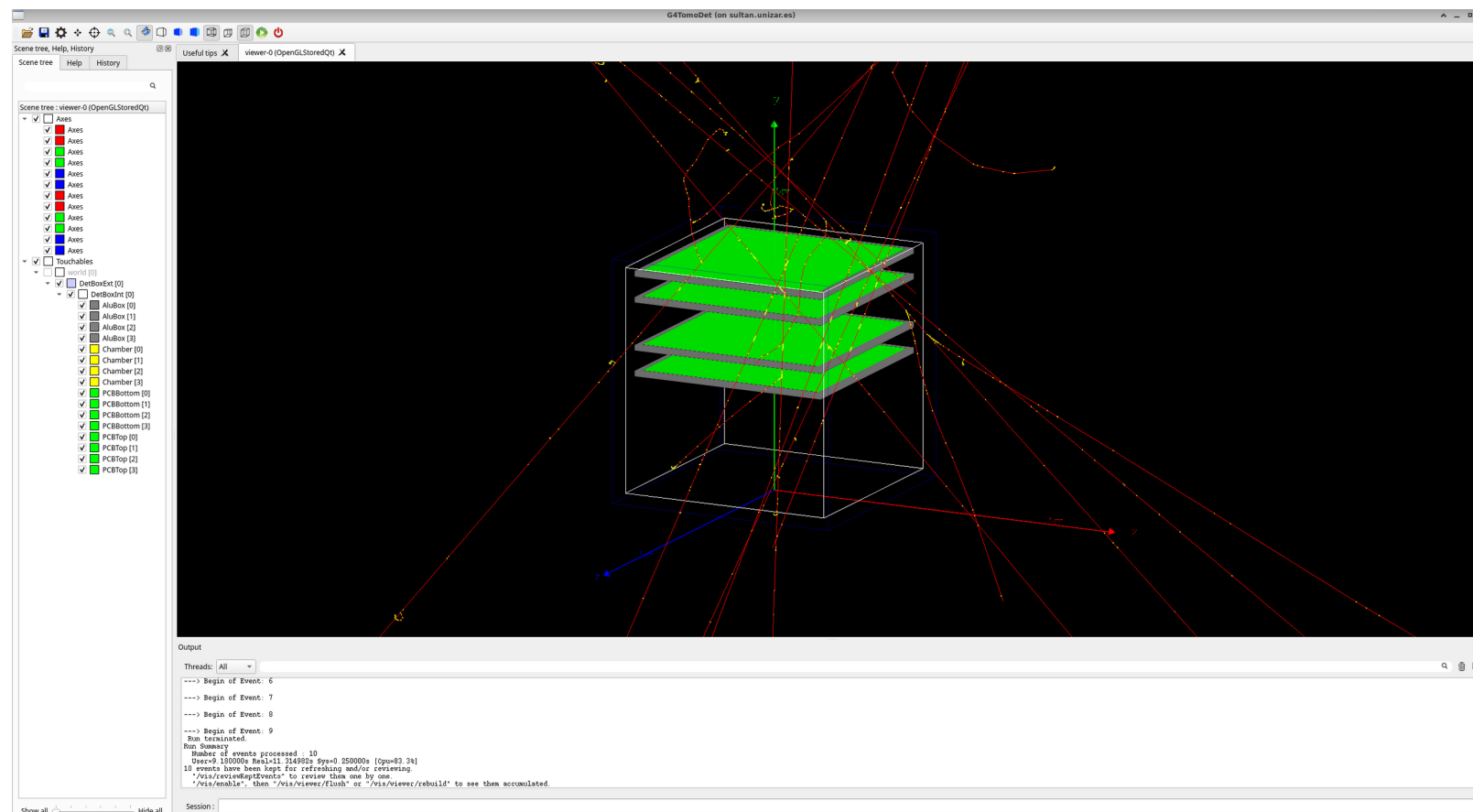
- Event information

- Energy deposit, position, particle ...

- Simulation is working @ sultan
(see Simulations_Manual_TFG2025)

Simulations – Geant4: G4TomoDet code
@ /home/hgomez/G4TomoDet_TFG2025

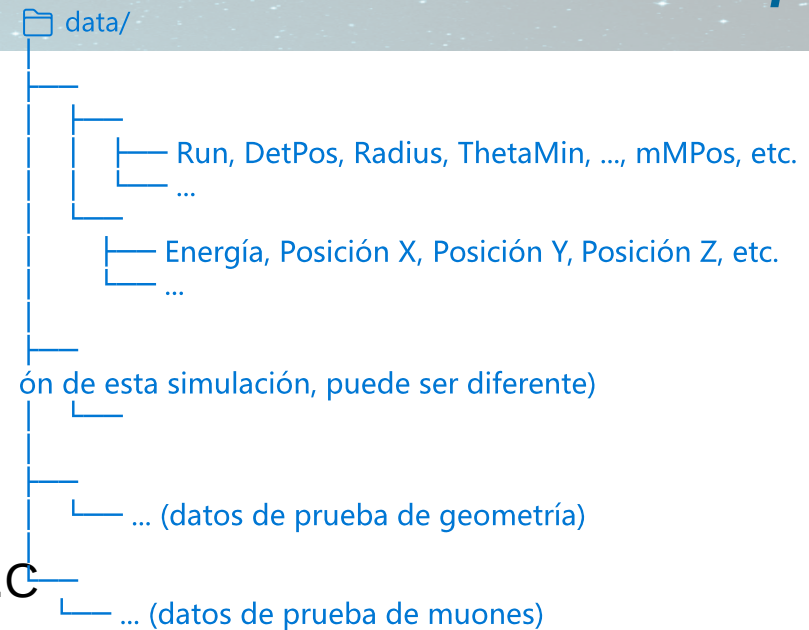
Output : G4TomoDet_*.root



Simulations – Geant4: G4TomoDet code
@ /home/hgomez/G4TomoDet_TFG2025

Output : G4TomoDet_*.root

- Two macros @scripts:
 - G4TomoDet_Data_Check_LightTree_General.C
 - Generate some control plots about the angular distribution of the detected muons
 - G4TomoDet_Data_Check_LightTree_Residual.C
 - Saves the X,Y,Z coordinates for each Micromegas of the detected muons
 - This is the starting point for your analysis work
- There is an example of all data files @data



en data: root -l --web=off -e 'new TBrowser();' NombreDelArchivo